input()

In [11]:
```
message=input("Tell me something and i will repeat back to you:")
print(message)
```

```
I learn python
hello
```

In [12]:
```
name=input("Please input your name:")
print("Hello,"+name.title())
```

```
Hello,Mica
Hello,Mica
```

input_prompt(提示语/多行)

In [13]:
```
prompt="If you tell us something,we can pernalize the messages you see."
prompt+="\nWhat's your name?"

name = input(prompt)
print("\nHello, " + name + "!")
```

```
Hello, mica!
Hello, mica!
```

input_numerical(注意用int() 或 float())

In [15]:
```
age = input("How old are you?")
int(age) > 20
```

Out[15]:
```
False
```

In [17]:
```
height = input("How tall are you,in inches?")
height = int(height)

if height >= 36:
    print("\nYou are tall enough to ride!")
else:
    print("Sorry,you are not old enough to ride.")
```

```
Sorry,you are not old enough to ride.
You are tall enough to ride!
```

input_modulo operator (%)

In [18]:
```
number=input("Please enter a number:")
number=int(number)

if number % 2 ==0:
    print("The number is Even(偶数).")
```

```
else:
    print("The number is Odd(奇数).")
```

```
The number is Odd(奇数).
The number is Even(偶数).
```

Excercise

In [19]:
```
# 7-1
car=input("What kind of rental car they would like?")
print("Let me see if I can find you a "+car)
```

```
Let me see if I can find you a Subaru
Let me see if I can find you a Subaru
```

In [20]:
```
# 7-2
group_num=input("How many people are in your dinner group ?")
group_num=int(group_num)

if group_num>8:
    print("You have to wait for a table.")
else:
    print("The table is ready.")
```

```
You have to wait for a table.
The table is ready.
```

In [21]:
```
# 7-3 multiple(倍数) of 10
num=input("Please enter a number:")
num=int(num)

if num % 10 ==0:
    print("The number is a multiple of 10.")
else:
    print("Sorry,the number is not a multiple of 10!")
```

```
Sorry,the number is not a multiple of 10!
The number is not a multiple of 10!
```

While loops

In [22]:
```
num=1
while num <=5:
    print(num)
    num+=1
```

```
1
2
3
4
5
```

while + prompt(提示语)

In [24]:
```
prompt="If you tell us something,we can pernalize the messages you see."
prompt+="\nEnter 'quit' to end the program."
```

```
message=""
while message !='quit':
    message=input(prompt)
    print(message + "\n")
```

```
my name is mica

i come from china

quit
```

In [26]:
```
# 薄弱题
# 电影名: Titanic / Kung Fu Hustle / Forrest Gump
prompt = "You can enter your favorite movie."
prompt += "\nEnter 'exit' to end the program."

message = ""  # 启动while
while message != 'exit':
    message = input(prompt) # 输入信息
    if message != 'exit':  # 可以避免输出“i like exit.”
        print("I like watching " + message.title() + "\n")
```

```
I like watching Forrest Gump
```

In [27]:
```
# 薄弱题练习1
prompt="You can enter your favorite food then i it will show on screen."
prompt+="\nPlease enter your favorite food."

info=''
while info != "exit":
    info=input(prompt)
    if info != "exit":
        print("My favorite food is " + info + ".\n")
```

```
My favorite food is beef.
```

In [29]:
```
prompt="You can enter your favorite food then i it will show on screen."
prompt+="\nWhat's your favorite book."

info=''
while info != 'q':
    info=input(prompt)
    if info != 'q':
        print('Your favorite book is ' + info + ".\n")
```

```
Your favorite book is Steven Jobs.
```

while + flag

In [30]:
```
prompt = "Enter 'exit' to end the program."
prompt += "\nWhat's your favorite movie."

flag=True  # 启动while
while flag:
    message=input(prompt)
```

```
    if message == 'exit':
        flag=False
    else:
        print(message.title() + ".\n")
```

```
Forrest Gump.
```

In [16]:
```
# while + flag 练习题
prompt="Enter 'exit' to end the program."
prompt+="\nWhat's your favorite subject?"

flag = True
while flag:
    info = input(prompt)
    if info == 'exit':
        flag=False
    else:
        print("\nMy favorite subject is " + info.title() + ".\n")
```

```
My favorite subject is Mathmatic.

My favorite subject is Data Science.
```

while + break

In [10]:
```
prompt = "\nPlease enter the name of a city you have visited:"
prompt += "\n(Enter 'quit' when you are finished.) "

while True:
    city=input(prompt)

    if city=='quit':
        break
    else:
        print("I've love to go to visit "+city.title())
```

```
I've love to go to visit Newyork
I've love to go to visit Kuala Lumpur
```

In [22]:
```
# while + break 练习题
prompt = "Enter 'q' to end the program."
prompt += "\nWhich country do you want to visit?"

while True:
    info = input(prompt)
    if info == 'q':
        break
    else:
        print("\nYou like to visit " + info +".\n")
```

```
You like to visit china.

You like to visit malaysia.
```

while + continue

In [11]:
```
num=0
while num<10:
    num+=1
    if num % 2==0:
        continue
    else:
        print(num)
```

```
1
3
5
7
9
```

Avoiding Infinite Loops(避免无限循环)

In [14]:
```
x=1
while x<5:
    print(x)
    x+=1
```

```
1
2
3
4
```

Excercise

In [31]:
```
# 7-4
prompt="We will add which topping you enter to present to you."
prompt+="\nEnter 'quit' if you want to finish your topping."

message=""
while message != 'quit':
    message=input(prompt)
    if message != 'quit':
        print("I will add "+message+" to your pizza.\n")
```

```
I will add beef to your pizza.

I will add chicken to your pizza.
```

while + if-elif + break

In [4]:
```
# 7-5 薄弱题
while True:
    info = input("\nPlease enter your age: ")
    if info.lower() == 'quit':
        break
```

```
    age = int(info)
    if age < 0 or age > 100:
        print("Invalid age!")
    elif age < 3:
        print("Ticket: Free")
    elif age <= 12:
        print("Ticket: $10")
    else:
        print("Ticket: $15")
```

```
Ticket: $10
Ticket: $15
Ticket: $15
```

In [27]:

```
# 7-5 练习题1
while True:
    info = input("\nPlese enter your height.")
    if info.lower() == 'q':
        break

    height = int(info)
    if height < 100:
        print("Ticket:FREE")
    elif 100 <= height <140:
        print("Ticket:$8")
    else:
        print("Ticket:$15")
```

```
Ticket:FREE
Ticket:$8
Ticket:$15
```

In [31]:

```
# 7-5 练习题2
prompt = "Enter 'q' to end the program."
prompt +="\nPlease enter your age."

while True:
    info = input(prompt)
    if info == 'q':
        break

    age = int(info)
    if age < 18:
        print("Entry fee : $5" )
    elif 18 <= age < 60:
        print("Entry fee : $12")
    else:
        print("Entry fee : $7")
```

```
Entry fee : $5

Entry fee : $12

Entry fee : $7
```

In [32]:

```
# 7-5 excercise
while True:
    info=input("\nPlease enter your height:")
    if info.lower()=='quit':
```

```
        break

    # unit:cm
    height=float(info)
    if height<0 or height>250:
        print("Invalid height!")
    elif height <80:
        print("Ticket Cost: FREE")
    elif 80<= height <120:
        print("Ticket Cost: $50")
    elif 120<= height <150:
        print("Ticket Cost: $80")
    else:
        print("Ticket Cost: $120")
```

```
Invalid height!
Ticket Cost: FREE
Ticket Cost: $50
```

while + if-elif + flag + continue

In [35]:
```
# 7-6 (1)using flag(active) 薄弱题
active=True
while active:
    info = input("\nPlease enter your age: ")
    if info.lower() == 'quit':
        active=False
        continue

    age = int(info)
    if age < 0 or age > 100:
        print("Invalid age!")
    elif age < 3:
        print("Ticket: Free")
    elif age <= 12:
        print("Ticket: $10")
    else:
        print("Ticket: $15")
```

```
Invalid age!
Ticket: Free
Ticket: $10
Ticket: $15
```

In [36]:
```
# 7-6 (2) while condition 薄弱题
info=""
while info.lower() != 'quit':
    info = input("\nPlease enter your age: ")
    if info.lower() == 'quit':
        continue

    age = int(info)
    if age < 0 or age > 100:
        print("Invalid age!")
    elif age < 3:
        print("Ticket: Free")
    elif age <= 12:
        print("Ticket: $10")
```

```
    else:
        print("Ticket: $15")
```

```
Ticket: Free
Ticket: $10
Ticket: $15
```

In [4]:
```
# 7-6 (3)using break  薄弱题
info=""
while info.lower() != 'quit':
    info = input("\nPlease enter your age: ")
    if info.lower() == 'quit':
        break

    age = int(info)
    if age < 0 or age > 100:
        print("Invalid age!")
    elif age < 3:
        print("Ticket: Free")
    elif age <= 12:
        print("Ticket: $10")
    else:
        print("Ticket: $15")
```

```
Ticket: Free
Ticket: $10
Ticket: $15
```

List to List moving

In [32]:
```
# 薄弱题
unconfirmed_users = ['alice', 'brian', 'candace']
confirmed_users = []

while unconfirmed_users:
    current_users=unconfirmed_users.pop()

    print("Verifying users:" + current_users.title())
    confirmed_users.append(current_users)

print("\nThe following users have been confirmed:")
for confirmed_user in confirmed_users:
    print(confirmed_user.title())
```

```
Verifying users:Candace
Verifying users:Brian
Verifying users:Alice

The following users have been confirmed:
Candace
Brian
Alice
```

In [44]:
```
# 练习题
pending_orders = ['black tea','orange juice','milk tee']
complete_orders = []

print("These orders have been confirmed:")
while pending_orders:
```

```
    current_orders = pending_orders.pop()
    print("-Confirmed Order:" + current_orders.title())
    complete_orders.append(current_orders)

print("\nThese orders have been completed:")
for complete_order in complete_orders:
    print("-" + complete_order.title())
```

```
These orders have been confirmed:
-Confirmed Order:Milk Tee
-Confirmed Order:Orange Juice
-Confirmed Order:Black Tea

These orders have been completed:
-Milk Tee
-Orange Juice
-Black Tea
```

Remove specific value from a list

In [20]:
```
# 薄弱题
pets = ['dog', 'cat', 'dog', 'goldfish', 'cat', 'rabbit', 'cat']
print(pets)

while 'cat' in pets:  # 注意cat加符号‘’，用while不是for，因为while可以判断，for用
    pets.remove('cat')
print(pets)
```

```
['dog', 'cat', 'dog', 'goldfish', 'cat', 'rabbit', 'cat']
['dog', 'dog', 'goldfish', 'rabbit']
```

In [46]:
```
# while + list + remove
fruits=['apple','orange','mango','banana','orange','watermelon','orange']
while 'orange' in fruits:
    fruits.remove('orange')
print(fruits)
```

```
['apple', 'mango', 'banana', 'watermelon']
```

Filling a Dictionary with User Input

In [26]:
```
# 薄弱题
responses={}
polling_active=True

while polling_active:
    name=input("\nWhat is your name?")
    response=input("Which mountain would you like to climb someday?")
    responses[name]=response

    repeat=input("Would you like to let another person respond? (yes/ no) ")
    if repeat=='no':
        polling_active=False

print("\n--- Poll Results ---")
for name,response in responses.items():
    print(name+" would you like to climb "+response+".")
```

```
--- Poll Results ---
Mica would you like to climb taishan.
Jack would you like to climb nanshan.
{'Mica': 'taishan', 'Jack': 'nanshan'}
```

In [54]:
```python
# while + dictionary + flag + input
colors = {}
flag = True

while flag:
    name = input("What's your name?")
    response = input("What's your favorite color?")
    colors[name] = response

    repeat = input("Do you want someone to answer?")
    if repeat.lower() == 'no':
        flag = False

print("\n--- The result as bellow: ---")
for k,v in colors.items():
    print(k.title() + "'s favorite color is " + v + ".")
```

```
--- The result as bellow: ---
Mica's favorite color is blue.
Jack's favorite color is yellow.
```

Excercise

In [2]:
```python
# while + list + flag + input
# 7-8 薄弱题
sandwich_orders=['tomato','beef','vegetable','lamb']
finished_sandwiches=[]

while sandwich_orders:
    current_sandwich=sandwich_orders.pop()
    print("I made your "+ current_sandwich+" sandwich")
    finished_sandwiches.append(current_sandwich)

print("\n---Sandwich that was made:---")
for finished_sandwich in finished_sandwiches:
    print(finished_sandwich + " sandwich was made")
```

```
I made your lamb sandwich
I made your vegetable sandwich
I made your beef sandwich
I made your tomato sandwich

---Sandwich that was made:---
lamb sandwich was made
vegetable sandwich was made
beef sandwich was made
tomato sandwich was made
```

In [6]:
```python
# 7-9
sandwich_orders=['tomato','pastrami','beef','pastrami','vegetable','lamb','pastr
finished_sandwiches=[]

while 'pastrami'in sandwich_orders:
    sandwich_orders.remove('pastrami')
```

```
print("The deli has run out of pastrami.\n\nWe only have bellow toppings:")
print(str(sandwich_orders)+"\n")

while sandwich_orders:
    current_sandwich=sandwich_orders.pop()
    print("I made your "+ current_sandwich+" sandwich.")
    finished_sandwiches.append(current_sandwich)

print("\n---Sandwich that was made:---")
for finished_sandwich in finished_sandwiches:
    print(finished_sandwich.title() + " sandwich was made.")
```

```
The deli has run out of pastrami.

We only have bellow toppings:
['tomato', 'beef', 'vegetable', 'lamb']

I made your lamb sandwich.
I made your vegetable sandwich.
I made your beef sandwich.
I made your tomato sandwich.

---Sandwich that was made:---
Lamb sandwich was made.
Vegetable sandwich was made.
Beef sandwich was made.
Tomato sandwich was made.
```

In [10]:

```
# 7-10
places={}
polling_active=True

while polling_active:
    name=input("\nWhat is your name?")
    response=input("If you could visit one place in the world, where would you g
    places[name]=response

    repeat=input("Would you like to let another person respond? (yes/ no) ")
    if repeat.lower()=='no':
        polling_active=False

print("\n--- Poll Results ---")
for name,response in places.items():
    print(name.title() + " would go to visit "+ response.title()+".")
```

```
--- Poll Results ---
Mica would go to visit Taishan.
Jack would go to visit Nanshan.
```

In [ ]:

In [ ]:

In [ ]: